

8. (1.25 points)

Two actuaries estimate the exponential claim size model parameters at the cost level of exposure year 5 and at the same basic limit via Sahasrabuddhe's approach to claim size modeling as follows:

<u>Actuary</u>	<u>Development Interval</u>				
	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>5</u>
A	50,000	120,000	165,000	195,000	205,000
B	25,000	65,000	95,000	115,000	120,000

All other assumptions made by the actuaries are the same.

Both actuaries use their claim size model to estimate cumulative development factors adjusted for trend and limit changes. They then compare these factors to the unadjusted cumulative development factors calculated using weighted-average link ratios that use all available development periods.

a. (0.75 point)

Identify which actuary's modeled development factors deviate further from the unadjusted development factors and explain the effect of different claim size parameters on modeled development patterns that causes such a result.

b. (0.5 point)

Briefly describe two other modeling choices that could have created larger differences between the modeled factors and the unadjusted factors, assuming there had been differences in the actuaries' other assumptions.

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EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 8	
TOTAL POINT VALUE: 1.25	LEARNING OBJECTIVE: A4: Estimate unpaid claims for various layers of claims.
SAMPLE ANSWERS	
Part a: 0.75 point	
<i>Sample Answer 1</i> Actuary A selected a larger θ for his exponential distribution. Therefore, a higher percentage of losses will be removed at a given limit, meaning varying limits will have a greater impact. Therefore, Actuary A will have large adjustment factors from the unadjusted LDFs. <i>Sample Answer 2</i> Actuary A's development factors will deviate further. With larger θ in the claim size model, the LEV will change more based on what limits you're comparing. The larger the θ, the more likely claims will be capped by the limit you're looking at since there is a higher potential for large losses with a larger mean in your claim size model. Then when looking at $LEV_{(limit)} / LEV_{(base)}$ you are like to see larger differences when your mean is larger in the claim size model. <i>Sample Answer 3</i> A will deviate further because of the larger claim size parameters, since the limiting will have a bigger effect on larger claims. Therefore, when adjusting for limit, it will be a bigger adjustment when claim sizes are larger.	
Part b: 0.5 point	
<i>Sample Answers (any two of which would earn full credit)</i> • Different AY trend assumptions. • Different CY trend assumptions. • If they made different trend assumptions which would cause the detrended limited means to differ. • They could have assumed that losses develop further out than 5 periods. • May have chosen a different claim size model distribution (e.g. gamma, pareto, etc.). • If either actuary made a simplifying assumption about R_j, the relationship between base layer and target layer, instead of calculating them from LEVs.	
EXAMINER'S REPORT	
Part a	
• Candidates were expected to know the interrelationships between parameters for forecasting LDFs • Candidates were expected to understand that the higher parameters/severities selected by Actuary A would influence the number of claims which may be capped at the selected limit in the model, therefore impacting the modeled LDFs and cause them to deviate further from the unadjusted LDFs.	

EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

- Due to the fact that we have not previously asked a similar type of question, it may have been a little more difficult for candidates to recognize what concept was being tested.
- Many candidates did not understand that the values provided for each Actuary were the underlying model parameters and not values used in a triangle to calculate LDFs.
- Many candidates calculated the range between development intervals 1 and 5 which did not have an impact on the deviation of LDFs.
- Some candidates correctly stated that Actuary A had higher parameters, but incorrectly reasoned that the higher parameters would lead to a higher variance in the exponential distribution, which is not a true cause of the larger deviation of the modeled LDFs.

Part b

- The candidate was expected to know the inputs that go into Sahasrabuddhe's approach to calculated LDFs.
- Additionally, though selecting a distribution other than the exponential was a valid answer, some candidates responded by stating two separate distributions and this did not receive full credit.
- Many candidates stated that changing the Basic Limit would impact the deviation of the modeled LDFs to the unadjusted LDFs. This is incorrect because it is based on a ratio of LEVs applied to data adjusted to the Basic Limit, and therefore the impact of the Basic Limit is cancelled out and does not influence the deviation between the LDFs.
- Some candidates tried to use theories/models from other portions of the syllabus, though the question specified that they were to respond in the context of Sahasrabuddhe's approach.